Check-In 01/15. *(True/False)* Making the u-substitution $u = x^2 + 1$ in $\displaystyle\int_0^1 \frac{5x}{x^2+1}\,dx$, we obtain $\displaystyle\int_0^1 \frac{5}{2u}\,du$.

Solution. The statement is *false*. One must always remember to change the limits when making a substitution in a definite integral. If we choose $u = x^2 + 1$, then $du = 2x\,dx$. If $x = 0$, then $u = 0^2 + 1 = 1$. If $x = 1$, then $u = 1^2 + 1 = 2$. But then…

$$\int_0^1 \frac{5x}{x^2+1}\,dx = 5\int_0^1 \frac{x}{x^2+1}\,dx = \frac{5}{2}\int_0^1 \frac{2x}{x^2+1}\,dx = \frac{5}{2}\int_1^2 \frac{du}{u}$$

We also have…

$$\frac{5}{2}\int_1^2 \frac{du}{u} = \frac{5}{2}\,\ln|u|\,\Big|_1^2 = \frac{5}{2}\,(\ln 2 - \ln 1) = \frac{5}{2}\,(\ln 2 - 0) = \frac{5}{2}\,\ln(2) = \ln(\sqrt{32}) \approx 1.73287$$

Check-In 01/20. *(True/False)* We should integrate $\displaystyle\int xe^{x^2}\,dx$ using integration-by-parts.

Solution. The statement is *false*. This certainly looks like an integration-by-parts integral—specifically, a tabular integral (because it is of the form polynomial times trig). The only possible choices for dv would be e^{x^2}, which cannot be integrated, or xe^{x^2}, which is the entire integral and we already 'don't know' how to integrate, or x. This last choices forces $u = e^{x^2}$. But then the new integral produced is $\displaystyle\int x^3 e^{x^2}\,dx$—which is even 'worse.' Instead, observe that the given integral can be found by u-substitution using $u = x^2$, so that $du = 2x\,dx$. But then…

$$\int xe^{x^2}\,dx = \frac{1}{2}\int 2xe^{x^2}\,dx = \frac{1}{2}\int e^u\,du = \frac{e^u}{2} + C = \frac{e^{x^2}}{2} + C$$

There is a heavy overlap between u-substitution and integration-by-parts in terms of what the integrals that require these methods 'look like.' One needs to be careful when determining when to use which method. Some integrals require both methods!

Check-In 01/22. *(True/False)* The integral $\displaystyle\int x^2 3^x\,dx$ can be integrated using tabular integration.

Solution. The statement is *true*. We anticipate that this might be an integration-by-parts tabular integral because it is of the form polynomial times exponential. Using LIATE, we choose $u = x^2$, which forces $dv = 3^x$. Note that $\displaystyle\int 3^x\,dx = \frac{3^x}{\ln 3} + C$. We then have…

u		dv
x^2	$+$	3^x
$2x$	$-$	$\frac{3^x}{\ln 3}$
2	$+$	$\frac{3^x}{(\ln 3)^2}$
0		$\frac{3^x}{(\ln 3)^3}$

Therefore, we have...

$$\begin{aligned}\int x^2 3^x \, dx &= \boxed{\frac{x^2(3^x)}{\ln 3} - \frac{2x(3^x)}{(\ln 3)^2} + \frac{2(3^x)}{(\ln 3)^3} + C} \\ &= \frac{3^x}{(\ln 3)^3}\big(x^2(\ln 3)^2 - 2x(\ln 3) + 2\big) + C \\ &= \frac{3^x\,\big(x^2(\ln 3)^2 - x(2\ln 3) + 2\big)}{(\ln 3)^3} + C \\ &= \frac{3^x\,\big(x^2(\ln 3)^2 - x\ln 9 + 2\big)}{(\ln 3)^3} + C\end{aligned}$$

Check-In 01/27. *(True/False)* To integrate $\int \tan^2 x \sec x \, dx$, one chooses $u = \sec x$.

Solution. The statement is *false*. Observe that if one chooses $u = \sec x$, we have $du = \sec x \tan x \, dx$. But writing $\int \tan x \cdot \sec x \tan x \, dx$, we have a remaining term of $\tan x$, which cannot be replaced with the trigonometric Pythagorean identity. Therefore, this substitution will not work. Similarly, choosing $u = \tan x$, we would have $du = \sec^2 x$—requiring two factors of $\sec x$ while the integrand only has one. To compute the given integral, observe...

$$\int \tan^2 x \sec x \, dx = \int \frac{\sin^2 x}{\cos^2 x} \cdot \frac{1}{\cos x} \, dx = \int \frac{\sin^2 x}{\cos^3 x} \, dx = \int \frac{1 - \cos^2 x}{\cos^3 x} \, dx = \int \big(\sec^3 x - \sec x\big) \, dx$$

Using the fact that $\int \sec^3 x \, dx = \frac{1}{2}\left(\sec x \tan x + \ln|\sec x + \tan x|\right) + C$ and $\int \sec x \, dx = \ln|\sec x + \tan x| + C$, we have...

$$\int \tan^2 x \sec x \, dx = \frac{\sec x \tan x - \ln|\sec x + \tan x|}{2} + C$$

Check-In 01/29. *(True/False)* Consider the integral $\int (x^2+4)^3\;dx$. One can recognize this is a trig. sub. because the "$a^2+b^2=c^2$–like" term x^2+4, and can then make the substitution $x=2\tan\theta$.

Solution. The statement is *true*. One can indeed make the substitution $x=2\tan\theta$, where $dx=2\sec^2\theta\;d\theta$, so that $x^2+4=4\sec^2\theta$. One then finds that the integral is $128\int\sec^8\theta\;d\theta$. However, there is a simpler way:

$$\int (x^2+4)^3\;dx = \int \big(x^6+12x^4+48x^2+64\big)\;dx = \frac{x^7}{7}+\frac{12x^5}{x}+16x^3+64x+C$$

Always carefully observe one's integrand. While one method may work, there may be other approaches which are simpler.

Check-In 02/03. *(True/False)* The decomposition of $\dfrac{5x-6}{x^2(x+3)}$ takes the form $\dfrac{A}{x}+\dfrac{Bx+C}{x^2}+\dfrac{D}{x+3}$.

Solution. The statement is *false*. While it is true that quadratic terms receive general linear terms in their numerator in a partial fraction decomposition, if the 'base term', i.e. the term being raised to a power, is linear, the top is only a constant. We should think of x^2 as $x^2=(x-0)^2$, so that the 'base term' is $x-0$, which is linear. The partial fraction decomposition should be. . .

$$\frac{5x-6}{x^2(x+3)}=\frac{A}{x}+\frac{B}{x^2}+\frac{C}{x+3}$$

Note that one could use the given decomposition. However, one would have redundant terms. One would find that $B=0$ and C would be the B from the correct partial fraction decomposition.

Check-In 02/05. *(True/False)* Heaviside's rule will allow you to find the full partial fraction decomposition for a rational function whose denominator consists of products of only linear terms, e.g. $\dfrac{6x-9}{(x+1)(x-3)^2}$.

Solution. The statement is *false*. Recall that Heaviside's method (the cover-up method) only allows one to find the coefficient associated with the highest power of linear terms. So, while it will not find any coefficients associated to quadratic terms, it will find some coefficients associated to linear terms—but only their highest power. We have a partial fraction decomposition of. . .

$$\frac{6x-9}{(x+1)(x-3)^2}=\frac{A}{x+1}+\frac{B}{x-3}+\frac{C}{(x-3)^2}$$

Heaviside's method will allow one to find A,C but not B. Therefore, it will not find the full partial fraction decomposition. [*Note.* There is a modification of Heaviside's method that will allow one to find all coefficients associated to linear terms: given the coefficient A_k for $\frac{A_k}{(x-a)^k}$, where $1\leq k\leq n$, we have $A_k=\dfrac{1}{(n-k)!}\lim\limits_{x\to a}\dfrac{d^{n-k}}{dx^{n-k}}\big((x-a)^nF(x)\big)$, where $F(x)$ is the original rational function. Of course, other methods, such as Keily's Method, are often simpler.]

Check-In 02/10. *(True/False)* The partial fraction decomposition of $\dfrac{9-x}{x^5+6x^4+9x^3} = \dfrac{9-x}{x^3(x^2+6x+9)}$ is $\dfrac{A}{x}+\dfrac{B}{x^2}+\dfrac{C}{x^3}+\dfrac{Ax+B}{x^2+6x+9}$.

Solution. The statement is *false*. Recall before performing a partial fraction decomposition, one needs to be sure that the degree of the numerator is strictly less than the degree of the denominator (or else one first needs to polynomial long divide) and that the denominator is factored into irreducible terms. Observe that $x^2+6x+9=(x+3)(x+3)=(x+3)^2$. Therefore, we have...

$$\frac{9-x}{x^5+6x^4+9x^3} = \frac{9-x}{x^3(x^2+6x+9)} = \frac{9-x}{x^3(x+3)^2} = \frac{A}{x}+\frac{B}{x^2}+\frac{C}{x^3}+\frac{D}{x+3}+\frac{E}{(x+3)^2}$$

Of course, one could use the originally given decomposition. Denoting the originally given decomposition with tildes, one would find $\tilde{A}=D$ and $\tilde{B}=3D+E$.

Check-In 02/19. *(True/False)* Because $\lim\limits_{n\to\infty} \dfrac{1}{n\ln(n)}=0$, the sequence $\displaystyle\sum_{n=2}^{\infty} \frac{1}{n\ln(n)}$ converges.

Solution. The statement is *false*. In fact, the given series diverges. One is making an incorrect inference from the Divergence Test. It is true that if $\lim\limits_{n\to\infty} a_n \neq 0$, then $\displaystyle\sum^{\infty} a_n$ diverges—this is the Divergence Test. However, the Divergence Test does not say that if $\lim\limits_{n\to\infty} a_n = 0$ that the series $\displaystyle\sum^{\infty} a_n$ converges. One cannot determine whether a series $\displaystyle\sum^{\infty} a_n$ converges or diverges from the fact that $\lim\limits_{n\to\infty} a_n=0$. Consider the series $\displaystyle\sum^{\infty}\frac{1}{n}$ (the Harmonic series) and $\displaystyle\sum^{\infty}\frac{1}{n^2}$. For both series, $\lim\limits_{n\to\infty}\dfrac{1}{n}=0$ and $\lim n\to\infty\dfrac{1}{n^2}=0$. However, the series $\displaystyle\sum^{\infty}\frac{1}{n}$ diverges (it is the Harmonic series), while the series $\displaystyle\sum^{\infty}\frac{1}{n^2}$ converges to $\frac{\pi^2}{6}$. So, while $\lim\limits_{n\to\infty} a_n=0$ in both cases, the series have different behaviors. The fact that $\lim\limits_{n\to\infty} a_n=0$ is never enough to determine the behavior of $\displaystyle\sum^{\infty} a_n$ without more information about the sequence $\{a_n\}$.

Check-In 02/24. *(True/False)* The series $\displaystyle\sum_{n=2}^{\infty} 4\left(\frac{9^{n+4}}{5^{2n}}\right)$ diverges.

Solution. The statement is *false*. Although it seems that for large n, $4\left(\frac{9^{n+4}}{5^{2n}}\right) \approx 4\cdot\frac{9^n}{5^n} \approx 4\left(\frac{9}{5}\right)^n$, which would diverge by the Geometric Series Test because $|r|=\frac{9}{5}>1$, this is *not* the case because the correct r has not been found. Given a geometric series, one needs to be sure to write it in the form $\displaystyle\sum^{\infty} ar^n$. That is, you need to find a common power n for all the exponential terms. For this series, we have...

$$\sum_{n=2}^{\infty} 4\left(\frac{9^{n+4}}{5^{2n}}\right) = \sum_{n=2}^{\infty} 4\left(\frac{9^n 9^4}{(5^2)^n}\right) = \sum_{n=2}^{\infty} (4 \cdot 9^4)\left(\frac{9^n}{25^n}\right) = \sum_{n=2}^{\infty} 26244\left(\frac{9}{25}\right)^n$$

Observe this series is geometric because it has the form $\sum^{\infty} ar^n$, where $a = 26244$ and $r = \frac{9}{25}$. Therefore, the series converges by the Geometric Series Test because $|r| = \frac{9}{25} < 1$. In fact, the series converges to $\frac{\text{first term}}{1-r} = \frac{2125764/625}{1-\frac{9}{25}} = \frac{531441}{100} \approx 5314.41$.

Check-In 02/26. *(True/False)* Because $\displaystyle\int_1^{\infty} \frac{x}{e^x}\,dx$ converges to $\frac{2}{e}$, the series $\displaystyle\sum_{n=1}^{\infty} \frac{n}{e^n}$ converges to $\frac{2}{e}$ by the integral test.

Solution. The statement is *false*. Let $f(x) = \frac{x}{e^x}$. We know that $f(x) > 0$ on $[1, \infty)$ because $x, e^x > 0$ on this interval. Because x, e^x are continuous on $[1, \infty)$, $f(x)$ is continuous because it is a quotient of continuous functions. Finally, $f(x)$ is decreasing on $[1, \infty)$ because $f'(x) = -e^{-x}(x-1) < 0$ on $[1, \infty)$. [Observe that for $x > 1$, $x - 1 > 0$, $e^x > 0$, so that $-e^x(x-1) < 0$.] Therefore by the Integral Test, we know that $\displaystyle\sum_{n=1}^{\infty} \frac{n}{e^n}$ converges if and only if $\displaystyle\int_1^{\infty} \frac{x}{e^x}\,dx$ converges. Using integration-by-parts, we know $\displaystyle\int \frac{x}{e^x}\,dx = -xe^{-x} - e^{-x} + C = -e^{-x}(x+1) + C = -\frac{x+1}{e^x} + C$. Therefore, we have...

$$\int_1^{\infty} \frac{x}{e^x}\,dx := \lim_{b \to \infty} \int_1^b \frac{x}{e^x}\,dx = \lim_{b \to \infty} -\frac{x+1}{e^x}\bigg|_1^b = \lim_{b \to \infty} -\frac{b+1}{e^b} - \left(-\frac{2}{e^1}\right) = 0 + \frac{2}{e} = \frac{2}{e}$$

By the Integral Test, because $\int_1^{\infty} \frac{x}{e^x}\,dx$ converges the series $\displaystyle\sum_{n=1}^{\infty} \frac{n}{e^n}$ converges. While the Integral Test guarantees that the expressions have the same behavior, it *does not* state that they have the same value. In fact, it should never be the case that they will be equal. So, while $\displaystyle\sum_{n=1}^{\infty} \frac{n}{e^n}$ converges, it does not converge to $\frac{2}{e}$. In fact, $\displaystyle\sum_{n=1}^{\infty} \frac{n}{e^n} = \frac{e}{(e-1)^2} \approx 0.9206735942$.

Check-In 03/03. *(True/False)* Because the series $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^5}$ converges by the p-test with $p = 5 > 1$, the series $\displaystyle\sum_{n=1}^{\infty} \frac{n^2}{n^5+1}$ converges by the Limit Comparison Test.

Solution. The statement is *false*. It is true that the series $\displaystyle\sum_{n=1}^{\infty} \frac{n^2}{n^5+1}$ converges. However, the stated reason is *not* the correct reasoning. One would need to compare to the series $\sum_{n=1}^{\infty} \frac{1}{n^3}$. We can see this from the fact that for sufficiently large n, $\frac{n^2}{n^5+1} \approx \frac{n^2}{n^5} = \frac{1}{n^3}$. Observe that...

$$\lim_{n\to\infty} \frac{\dfrac{n^2}{n^5+1}}{\dfrac{1}{n^3}} = \lim_{n\to\infty} \frac{n^2}{n^5+1} \cdot \frac{n^3}{1} = \lim_{n\to\infty} \frac{n^5}{n^5+1} = 1$$

Because this limit is finite and nonzero, we know by the Limit Comparison Test that the series $\sum_{n=1}^{\infty} \frac{n^2}{n^5+1}$ and $\sum_{n=1}^{\infty} \frac{1}{n^3}$ either both converge or both diverge. We know that the series $\sum_{n=1}^{\infty} \frac{1}{n^3}$ converges by the p-test because $p = 3 > 1$. Therefore, it must be that the series $\sum_{n=1}^{\infty} \frac{n^2}{n^5+1}$ converges.

Check-In 03/05. *(True/False)* The series $\sum_{n=1}^{\infty} (-1)^n \frac{|\arctan\left(\csc(\sqrt[3]{n})\right)|\sqrt[8]{n^\pi+5}}{n^7+\ln(n)+15}$ converges.

Solution. The statement is *true*. Observe that $|\arctan\left(\csc(\sqrt[3]{n})\right)|$, $\sqrt[8]{n^\pi+5}, n^7+\ln(n)+15 > 0$ on $[1,\infty)$. Therefore because of the $(-1)^n$ term, the series is alternating. We know that $|\arctan\left(\csc(\sqrt[3]{n})\right)|$ is bounded because $\arctan x$ is bounded. We know that for large n, $\sqrt[8]{n^\pi+5} \approx \sqrt[8]{n^\pi} = n^{\pi/8} \approx n^{0.393}$. Furthermore, we know for sufficiently large n that $n^7 + \ln n + 15 \approx n^7$. So for sufficiently large n, $\frac{|\arctan\left(\csc(\sqrt[3]{n})\right)|\sqrt[8]{n^\pi+5}}{n^7+\ln(n)+15} \approx \frac{n^{0.392}}{n^7}$. But then $\lim_{n\to\infty} \frac{|\arctan\left(\csc(\sqrt[3]{n})\right)|\sqrt[8]{n^\pi+5}}{n^7+\ln(n)+15} = 0$. Furthermore for sufficiently large n, we know that $\left\{\frac{|\arctan\left(\csc(\sqrt[3]{n})\right)|\sqrt[8]{n^\pi+5}}{n^7+\ln(n)+15}\right\}$ is a decreasing sequence. Therefore, the series $\sum_{n=1}^{\infty} (-1)^n \frac{|\arctan\left(\csc(\sqrt[3]{n})\right)|\sqrt[8]{n^\pi+5}}{n^7+\ln(n)+15}$ converges by the Alternating Series Test. Furthermore, adding the first thirty terms of the series, we estimate the series has a sum of 0.0766257845. By the Alternating Series Estimation Theorem, the error in this estimation will at most be the magnitude of the 31st term of the summand, which is approximately $2.2 \cdot 10^{-10}$.

Check-In 03/17. *(True/False)* Applying the Root Test to $\sum_{n=1}^{\infty} \left(\frac{1}{n}\right)^n$, we have $\left|\left(\frac{1}{n}\right)^n\right|^{1/n} = \frac{1}{n}$. Because $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges, we then know that $\sum_{n=1}^{\infty} \left(\frac{1}{n}\right)^n$ diverges by the Root Test.

Solution. The statement is *false*. Consider the infinite series $\sum^{\infty} a_n$ and let $L := \lim_{n\to\infty} |a_n|^{1/n}$. Recall that the Root Test states that $\sum^{\infty} a_n$ converges absolutely if $L < 1$, diverges if $L = 1$, and is inconclusive if $L = 1$. Observe for this series...

$$\lim_{n\to\infty} \left| \left(\frac{1}{n}\right)^n \right|^{1/n} = \lim_{n\to\infty} \frac{1}{n} = 0 < 1$$

Because this limit is less than 1, we know the series $\displaystyle\sum_{n=1}^{\infty} \left(\frac{1}{n}\right)^n$ is absolutely convergent.

Check-In 03/26. *(True/False)* There is a power series which has an interval of convergence given by $x < -5$ and $x > 5$.

Solution. The statement is *false*. First of all, the x-values for which $x < -5$ and $x > 5$ is not an interval. Furthermore, let $\displaystyle\sum^{\infty} a_n(x-c)^n$ be a power series with center c. The only possibilities for the interval of convergence for a power series are just the center, $\{c\}$, a finite interval of radius R centered about the center, e.g. $(c-R, c+R), [c-R, c+R), (c-R, c+R]$, or $[c-R, c+R]$, or the entire real line $(-\infty, \infty)$. So, there cannot be a power series with 'interval of convergence' given by $x < -5$ and $x > 5$.

Check-In 03/31. *(True/False)* If a power series converges on $[3, 19)$, then the radius of convergence must be 8 and the center must be $x = 11$.

Solution. The statement is *true*. The center of this interval of convergence is clearly $\frac{a+b}{2} = \frac{3+19}{2} = \frac{22}{2} = 11$. Furthermore, the radius of this interval of convergence is $R = \frac{b-a}{2} = \frac{19-3}{2} = \frac{16}{2} = 8$.

Check-In 04/02. *(True/False)* The Maclaurin series for $f(x)$ is $\displaystyle\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}\, x^n$.

Solution. The statement is *true*. Assuming that a function $f(x)$ is infinitely differentiable, the Taylor series for $f(x)$ centered at $x = c$ is given by $T(x) = \displaystyle\sum_{n=0}^{\infty} \frac{f^{(n)}(c)}{n!}\,(x-c)^n$. A Maclaurin series is simply a Taylor series with a center of $c = 0$. But then the Maclaurin series for $f(x)$ is $T(x) = \displaystyle\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}\,(x-0)^n = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}\, x^n$.

Check-In 04/07. *(True/False)* We have $\displaystyle\sum_{n=0}^{\infty} (-1)^n\, \frac{\pi^{2n}}{(2n)!} = -1$.

Solution. The statement is *true*. We know the Taylor series for $\cos x$ centered at $x = 0$ is

$\sum_{n=0}^{\infty}(-1)^n \frac{x^{2n}}{(2n)!}$. Furthermore, this Taylor series converges to $\cos x$ for all $x \in (-\infty, \infty)$. So, we can evaluate this Taylor series at $x = \pi$. But then...

$$\sum_{n=0}^{\infty}(-1)^n \frac{\pi^{2n}}{(2n)!} = \sum_{n=0}^{\infty}(-1)^n \frac{x^{2n}}{(2n)!}\bigg|_{x=\pi} = \cos(x)\bigg|_{x=\pi} = \cos(\pi) = -1$$

Check-In 04/09. *(True/False)* The 3rd degree Taylor polynomial, $T_3(x)$, for e^x centered at $x = 0$ is $1 + x + \frac{x^2}{2!} + \frac{x^3}{3!}$. The Taylor remainder term for this polynomial in approximating e^x 'near' $x = 0$ is $R_3(x) = \frac{x^4}{4!}$.

Solution. The statement is *false*. The Taylor Remainder Theorem (or Lagrange Remainder Theorem) states that the remainder (error) in approximating $f(x)$ using $T_n(x)$ (centered at c) is $R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!}\,(x-c)^{n+1}$, where c is some real number between x and c. That is, the remainder term is the 'next term' of the Taylor series but with the derivative instead evaluated at some other value between x and c—not the center! So for $f(x) = e^x$ and $T_3(x)$ (centered at 0), the remainder term would be $R_3(x) = \frac{f^{(4)}(c)}{4!}\,(x-0)^4 = \frac{f^{(4)}(c)}{4!}\,x^4$. Because $f^{(n)}(x) = e^x$ for all n, we have $R_3(x) = \frac{e^c}{4!}\,x^4$. Notice if the fourth derivative *had* been evaluated at the center 0, we would have had $\frac{e^0}{4!}\,x^4 = \frac{1}{4!}\,x^4 = \frac{x^4}{4!}$, which is the exact next term of the Taylor series.